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MCCS-Det: A Small-Scale Smoke Detection with YOLO26 Enhanced by Multi-Scale Context Compensation and a Self-Collaborative Head

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***Abstract*—Fire early warning is crucial for reducing disaster losses at the initial stage. As an important visual cue in early fire scenarios, smoke often appears with small scale, semi-transparency, and loose morphology, which makes it prone to missed detection under low-contrast conditions. To address this issue, this paper proposes MCCS-Det, an early small-scale smoke detection network based on YOLO26. The proposed method introduces a Multi-Scale Context Compensation Module (MCCM) to enhance weak smoke features through parallel max-pooling, average-pooling, and residual reinjection, and further designs a Self-Collaborative Decoupled Detection Head (SCHead) to improve cross-layer alignment between classification and localization features. Experimental results on the curated FASDD_CV dataset show that MCCS-Det achieves an inference speed of 1143.52 FPS, while the precision and recall for small-scale smoke reach 0.672 and 0.690, respectively, demonstrating superior accuracy and real-time performance over existing mainstream methods.**

***Index Terms*—smoke detection, small object detection, multi-scale context compensation**

I. Introduction

Fire is a highly destructive emergency and remains one of the most hazardous disasters to human society and the natural environment [1]. Because fire spread is strongly time-sensitive, early detection is essential for effective warning and intervention [2]. As a key visual cue in the initial stage of fire, smoke usually appears earlier than flames and therefore has broad application value in early fire warning.

In the early stage of a fire, smoke generally exhibits thin, semi-transparent, and loosely structured visual characteristics with gradual boundary transitions [3]. Small-scale smoke occupies only a limited number of pixels and is often coupled with complex backgrounds, such as sky, clouds, industrial steam, road dust, and tree canopy textures. Under low-contrast conditions, these factors make early smoke targets more prone to missed detection [4]. Therefore, improving the robustness of early small-scale smoke detection has become an important issue in computer vision and safety monitoring.

Existing studies have explored this problem from different perspectives. Yuan *et al.* proposed a wave-shaped deep neural network to alleviate feature loss in smoke density estimation and localization through multi-scale feature extraction [5]. Lin *et al.* introduced the 3DVSD network, which employs end-to-end 3D convolutions to improve dynamic smoke recognition in video streams [6]. Wang *et al.* designed SSmokeDet to enhance the representation of tiny smoke through a dedicated SmallNet backbone and a Multi-SPP module [4]. More recently, YOLO26 has shown strong small-object detection capability with very low latency by adopting end-to-end NMS-free inference, Small-Target Aware Label Assignment (STAL), and the MuSGD optimizer [7].

The main contributions of MCCS-Det are summarized as follows:

- We propose a Multi-Scale Context Compensation Module (MCCM) to strengthen deep feature representation for tiny and semi-transparent smoke through multi-scale pooling and residual reinjection.
- We design a Self-Collaborative Decoupled Detection Head (SCHead) to improve the consistency between classification and localization, thereby enhancing the detection robustness of small-scale smoke in complex scenes.

II. Related Work

Early smoke detection technologies primarily relied on hand-crafted feature operators, achieving identification by extracting smoke color models, texture characteristics, and dynamic motion vectors. These traditional methods demonstrated a degree of effectiveness in specific, constrained industrial environments. However, when faced with open and complex scenes involving intense lighting fluctuations, occlusions, or smoke-like interferents, their feature representation exhibits

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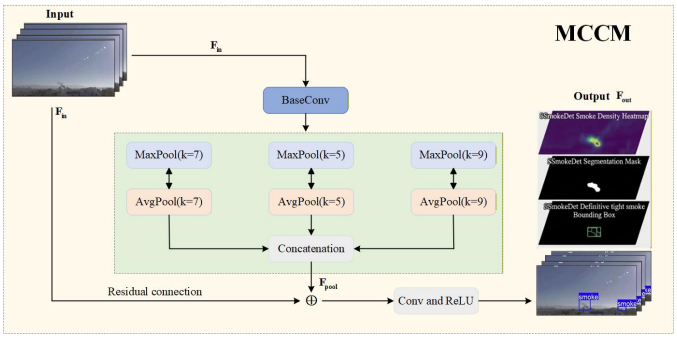


Fig. 1. Overview of the proposed Multi-Scale Context Compensation Module (MCCM) for small smoke detection.

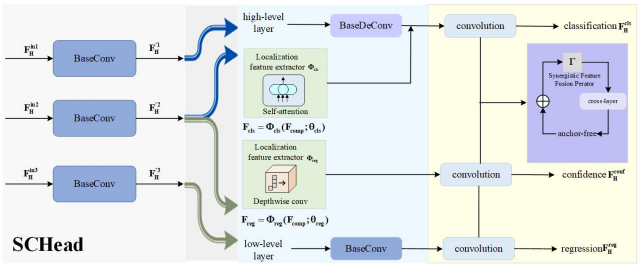


Fig. 2. The architecture of the proposed Self-collaborative Decoupled Head (SCHead).

significant limitations, making it difficult for false alarm and missed detection rates to meet practical requirements [8].

In recent years, the field of computer vision has developed rapidly. Convolutional Neural Networks (CNNs), with their superior local feature extraction and translational invariance, have become the fundamental framework for visual representation learning [9]. With the advancement of deep learning technology, many scholars have begun utilizing CNNs to automatically extract high-level semantic features, significantly enhancing the stability of smoke detection in complex environments [10]. Meanwhile, feature fusion strategies can effectively address the multi-scale characteristics of smoke [11].

III. Method

A. Multi-Scale Context Compensation Module

To address the weak representation of early-stage smoke with high transparency and blurred boundaries, MCCM is embedded into the deep feature extraction stage, as shown in Fig. 1. The module adopts parallel MaxPool and AvgPool operations under multiple receptive fields to aggregate contextual information. For the predefined pooling kernel set $\mathcal{K} = \{k_1, k_2, \ldots, k_n\}$, the aggregated feature is expressed as

$$F_{pool} = \mathrm{Concat}\left(\bigcup_{k_i \in \mathcal{K}} \{P_{\max}(F_{in}, k_i), P_{\mathrm{avg}}(F_{in}, k_i)\}\right) \quad (1)$$

To preserve the original feature details while enhancing contextual representation, the aggregated feature is fused with the input feature through residual reinjection. After channel fusion and dimensional alignment by a 1×1 convolution, the output feature is defined as

$$F_{out} = \mathrm{Conv}_{1\times1}(F_{pool}) + F_{in}. \quad (2)$$

Although MCCM is designed to enhance weak smoke features by aggregating contextual cues under multiple receptive fields, its effectiveness is related to the choice of the kernel set K. A too limited kernel configuration cannot sufficiently capture the gradual boundary transition and local saliency of early smoke, whereas excessively large kernels may over-smooth sparse smoke responses and introduce redundant background information. Therefore, the goal of MCCM is not to simply increase the number or size of pooling branches, but to establish a moderate and complementary receptive-field combination for weak smoke perception. As further analyzed in Section IV-C, the configuration $K = \{3, 5, 7\}$ provides the best trade-off between detail preservation and context compensation in our setting.

B. Self-Collaborative Decoupled Detection Head

To reduce the feature conflict between classification and localization in traditional smoke detection heads, a Self-Collaborative Decoupled Head (SCHead) is constructed, as shown in Fig. 2. The head separates the prediction process into a classification branch and a localization branch, and the decoupled mapping is formulated as

$$F_{cls} = \Phi_{cls}(F_{comp}; \theta_{cls}), \quad F_{reg} = \Phi_{reg}(F_{comp}; \theta_{reg}). \quad (3)$$

where Φ_{cls} and Φ_{reg} denote the nonlinear transformation functions for classification and localization, respectively. To enhance cross-branch interaction, a self-collaborative fusion mechanism is introduced to align semantic and geometric information. The final prediction output is defined as

$$P = \{P_{cls}, P_{reg}\} = \Gamma(F_{cls} \odot F_{reg}) + \Psi(F_{comp}). \quad (4)$$

where $\odot$ denotes the feature-level collaborative enhancement operator, and Ψ denotes the residual adjustment mapping.

In SCHead, the self-collaborative fusion is designed as a lightweight feature interaction rather than a heavy attention-based module. Specifically, the classification feature F_{cls} and localization feature F_{reg} are generated by two decoupled branches, and their element-wise collaborative response is used to highlight spatial positions jointly supported by semantic and geometric cues. Meanwhile, the residual adjustment term $\Psi(F_{comp})$ preserves the complementary representation and suppresses unstable activations caused by branch disagreement. As a result, SCHead improves the consistency between classification and localization while introducing only limited computational overhead, thereby maintaining the real-time advantage of YOLO26.

C. Loss Function

The loss function of MCCS-Det consists of three parts: classification loss L_{cl}, bounding box regression loss L_{iou}, and auxiliary coordinate regression loss L_1. The total loss is defined as

$$L = \lambda_{cl} L_{cl} + \lambda_{iou} L_{iou} + \lambda_1 L_1. \quad (5)$$

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TABLE I
COMPARISON OF DETECTION ACCURACY AND INFERENCE EFFICIENCY BETWEEN MCCS-DET AND OTHER MAINSTREAM METHODS

Method	AP_S	AP_M	AP_L	AR_S	AR_M	AR_L	AP	AR	FPS
Faster R-CNN [13]	0.368	0.560	0.601	0.521	0.640	0.660	0.599	0.662	7.790
SSD [14]	0.530	0.637	0.717	0.621	0.688	0.625	0.650	0.697	38.375
YOLOX [15]	0.471	0.652	0.736	0.580	0.699	0.654	0.725	0.628	232.081
STCNet [16]	0.383	0.484	0.601	0.520	0.591	0.662	0.582	0.654	132.568
SSmokeDet [4]	0.543	0.652	0.756	0.629	0.700	0.770	0.736	0.712	204.635
YOLO26 [7]	0.602	0.771	0.928	0.664	0.664	**0.825**	**0.919**	0.815	**1195.56**
MCCS-Det	**0.627**	**0.802**	**0.938**	**0.690**	**0.717**	**0.825**	0.914	**0.837**	1143.52

where λ_{cl}, λ_{iou}, and λ_1 are tunable hyperparameters.

For classification, we adopt the Varifocal Loss (VFL) from VarifocalNet, which is formulated as

$$L_{cl}(p,q) = \begin{cases} -q\left(q\log(p) + (1-q)\log(1-p)\right), & q > 0, \\ -\alpha p^{\gamma}\log(1-p), & q = 0. \end{cases} \tag{6}$$

where p denotes the predicted classification score, q denotes the IoU between the predicted box and the ground-truth box, and α is the modulation factor for negative samples. For localization, the regression loss combines an IoU-based term and a direct coordinate supervision term. The IoU regression loss is

$$L_{iou} = 1 - \mathrm{IoU} + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha_{iou} v, \tag{7}$$

where $\rho^2(b, b^{gt})$ denotes the Euclidean distance between the centers of the predicted box and the ground-truth box, c is the diagonal length of the minimum enclosing box, and α_{iou} is determined by the current IoU.

The auxiliary coordinate regression loss is

$$L_1 = \frac{1}{n}\sum_{i=1}^{n}\left|b_i - b_i^{gt}\right|. \tag{8}$$

where b_i and b_i^{gt} denote the predicted and ground-truth box coordinates, respectively.

IV. EXPERIMENT

A. Database

To ensure the generalization ability and detection reliability of the model in complex real-world environments, we adopt the FASDD_CV subset of FASDD [12] as the core data source. After strict preprocessing and filtering, low-quality images and background samples irrelevant to the detection targets are removed, and a curated dataset containing 14,768 high-quality images is finally obtained. Among them, 70%, 20%, and 10% are randomly divided into the training, test, and validation sets, respectively.

Furthermore, to further analyze the sensitivity of the model to smoke targets at different detection distances and scales, smoke targets are categorized according to the pixel area they occupy in the image plane. Specifically, small-scale smoke ranges from 0 to 32×32, medium-scale smoke ranges from 32×32 to 96×96, and large-scale smoke is greater than 96×96.

B. Comparison Experiment

We selected several classical object detection and smoke detection methods as baselines, and the results are summarized in Table I. Traditional detectors such as Faster R-CNN and SSD show limited suitability for smoke detection due to either low inference efficiency or insufficient overall accuracy. Although YOLOX achieves high speed, its relatively low AR indicates a higher missed detection rate, while STCNet performs poorly on small- and medium-scale smoke, with AP_S and AP_M of only 0.383 and 0.484, respectively. SSmokeDet improves the detection of small smoke targets, achieving $AP = 0.736$ and $AR = 0.712$, which confirms the effectiveness of smoke-oriented design. YOLO26 provides a strong baseline with excellent real-time performance and strong small-object perception. On this basis, MCCS-Det further improves smoke-specific detection performance. Although its overall AP is 0.005 lower than that of YOLO26, it achieves the highest AP_S, AP_M, and AP_L values and the best overall AR, indicating stronger detection and recall capability for small- and medium-scale smoke while maintaining high inference speed.

C. Ablation Study

TABLE II
ABLATION STUDY OF THE CORE COMPONENTS IN MCCS-DET

Method	AP_S	AP	AR_S	AR
YOLO26	0.602	**0.919**	0.664	0.815
YOLO26+MCCM	0.619	0.900	0.674	0.798
YOLO26+SCHead	0.606	0.914	0.663	0.811
YOLO26+SCHead+MCCM	**0.627**	0.914	**0.690**	**0.837**

To verify the effectiveness of MCCS-Det, we conducted a series of ablation experiments on the curated FASDD dataset, as shown in Table II. Using YOLO26 as the baseline, after introducing the MCCM, AP_S increased from 0.602 to 0.619. After further adding the self-collaborative decoupled prediction head, AP_S was further improved to 0.627, which proves that the introduction of MCCM and SCHead is effective in

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TABLE III
PERFORMANCE COMPARISON OF DIFFERENT KERNEL SET CONFIGURATIONS IN MCCM

Number	Kernel Sets K	AP_S	AP
1	$\{3\}$	0.505	0.867
2	$\{3,5\}$	0.614	0.879
3	$\{3,5,7\}$	**0.627**	**0.914**
4	$\{5,9,13\}$	0.616	0.903
5	$\{3,5,7,9,11\}$	0.576	0.857

enhancing the representation ability for thin, semi-transparent, and small-scale smoke targets.

In addition, in the scale configuration experiments of MCCM, we adopted a series of fine-grained combinations to find the optimal performance, as shown in Table III. When $K = \{3\}$ or $K = \{3,5\}$, due to the overly single receptive field sampling, the model is limited in its ability to aggregate smoke edge saliency and background textures, resulting in a relatively low AP_S. When $K = \{3,5,7\}$, the model achieves the best results. As the scale is further increased, an excessively large receptive field instead causes tiny smoke features to be diluted by complex background information, resulting in a certain decline in accuracy.

V. CONCLUSION AND FUTURE WORK

To address the challenges of small target scales and sparse features in early smoke detection, this paper proposes MCCS-Det, a high-speed detector built upon YOLO26. The framework introduces two core components: the Multi-Scale Context Compensation Module (MCCM), which strengthens weak smoke features via multi-scale pooling and residual reinjection (achieving optimal performance at $K = \{3,5,7\}$), and the Self-Collaborative Decoupled Detection Head (SC-Head), which aligns classification and localization through a lightweight reorganization mechanism. Experimental results on the FASDD dataset demonstrate competitive accuracy and real-time performance, with AP_S increasing from 0.602 to 0.627 and AR_S reaching 0.690. Despite these gains, the model exhibits sensitivity to pooling-scale configurations, and its robustness under extreme resolution shifts or intense smoke-like interference requires further validation. Future research will explore adaptive kernel selection, resolution-aware compensation, and temporal modeling to better handle dynamic smoke diffusion and complex deployment environments.

REFERENCES

[1] V. Loboichenko *et al.*, "Review of advances in fire extinguishing based on computer vision applications: Methods, challenges, and future directions," *Sensors*, vol. 25, no. 20, Art. no. 6399, 2025.

[2] S. Chaturvedi, P. Khanna, and A. Ojha, "A survey on vision-based outdoor smoke detection techniques for environmental safety," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 185, pp. 158–187, 2022.

[3] Z. Wang, L. Lei, T. Li, X. Zu, and P. Shi, "RT-DETR-Smoke: A real-time transformer for forest smoke detection," *Fire*, vol. 8, no. 5, Art. no. 170, 2025.

[4] J. Wang, L. Wang, R. Zhang, X. Li, and B. Fan, "SSmokeDet: A novel network dedicated to small-scale smoke detection," *Engineering Applications of Artificial Intelligence*, vol. 145, Art. no. 110092, 2025.

[5] F. Yuan, L. Zhang, X. Xia, Q. Huang, and X. Li, "A wave-shaped deep neural network for smoke density estimation," *IEEE Transactions on Image Processing*, vol. 29, pp. 2301–2313, 2019.

[6] Y. Huo, Q. Zhang, Y. Zhang, J. Zhu, and J. Wang, "3DVSD: An end-to-end 3D convolutional object detection network for video smoke detection," *Fire Safety Journal*, vol. 134, Art. no. 103690, 2022.

[7] R. Sapkota, R. H. Cheppally, A. Sharda, and M. Karkee, "YOLO26: Key architectural enhancements and performance benchmarking for real-time object detection," *arXiv preprint arXiv:2509.25164*, 2025.

[8] K. Gu, Z. Xia, J. Qiao, and W. Lin, "Deep dual-channel neural network for image-based smoke detection," *IEEE Transactions on Multimedia*, vol. 22, no. 2, pp. 311–323, 2019.

[9] T. Wang, J. Hong, Y. Han, G. Zhang, S. Chen, T. Dong, Y. Yang, and H. Ruan, "AOSVSSNet: Attention-guided optical satellite video smoke segmentation network," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 15, pp. 8552–8566, 2022.

[10] Y. Wang, Y. Wang, Z. A. Khan, A. Huang, and J. Sang, "Multi-level feature fusion networks for smoke recognition in remote sensing imagery," *Neural Networks*, vol. 184, Art. no. 107112, 2025.

[11] W. Wang *et al.*, "InternImage: Exploring large-scale vision foundation models with deformable convolutions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 14408–14419.

[12] M. Wang, P. Yue, L. Jiang, D. Yu, T. Tuo, and J. Li, "An open flame and smoke detection dataset for deep learning in remote sensing based fire detection," *Geospatial Information Science*, vol. 28, no. 2, pp. 511–526, 2025.

[13] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *Advances in Neural Information Processing Systems*, vol. 28, 2015.

[14] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 21–37.

[15] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," *arXiv preprint arXiv:2107.08430*, 2021.

[16] Y. Cao, Q. Tang, and X. Lu, "STCNet: Spatiotemporal cross network for industrial smoke detection," *Multimedia Tools and Applications*, vol. 81, no. 7, pp. 10261–10277, 2022.

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